

Design and Fabrication of Automatic Single Axis Solar Tracker for Solar Panel

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Abstract— Sun energy is a considered to be one of most promising source to address the world energy crises. Photovoltaic cell is one of the prominent sources of energy. The most important factors that affect the efficiency of solar cells are cell temperature, maximum power point tracking (MPPT) and energy conversion efficiency. The optimization of these factors improves solar cells efficiency for more reliable applications. This study associated by means of the designing and manufacturing process of single axis tracker device by using photo voltaic conversion panels. This solar tracker system assures the conversion optimization of electricity from the source of sun energy by proper using of orientated photovoltaic panel in a manner conforming to the actual direction of sun. Experimental system based on the function of DC motor, which is controlled by dedicated drive astutely so as to move the PV panel in accordance with the signals be given as of effective light sensors. It has been found that at the same sunny day the energy output of solar panel with tracking system was 1742.88Wh and with non tracking system was 829.6Wh and

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I. INTRODUCTION

Due to the dwindling and pollution of fossil fuels, many researchers have got attention to the advancement of renewable energy [1, 2]. Solar energy is the best option now a day's using worldwide. In 2016, the maximum consumption of solar energy has been observed. Compiled data from various official sources as shown in Fig 1 [3]. Solar cell is the main source that converts the light energy in to electricity [4].

The quantity, quality and time availability of solar radiations are the three important parameters which effects the conversion of solar energy [5]. Automatic solar tracking system is the possible approach to maximize the solar cell efficiency [6]. Tracker system tracks the location of sun hence it increases the input of sun radiation. Fig. 2 shows the fixed photovoltaic and tracking photovoltaic system [7-9]. The novelty of tracking system of sun energy is to enhance the tilt angle for obtaining the maximum available radiations [10].

The aim of this research is to fabricate a solar tracking system for photovoltaic cell to harvest the maximum efficiency. To analyse critical factors that affect the efficiency of solar photovoltaic cell by rotating the solar panel from east to west

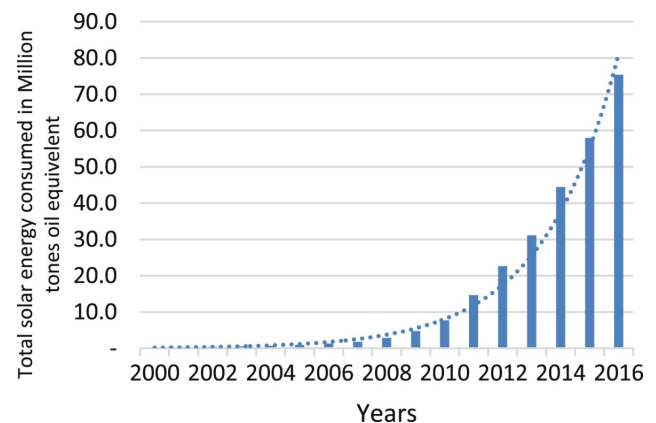


Fig. 1 Total sun energy consumed in Million tonnes oil equivalent

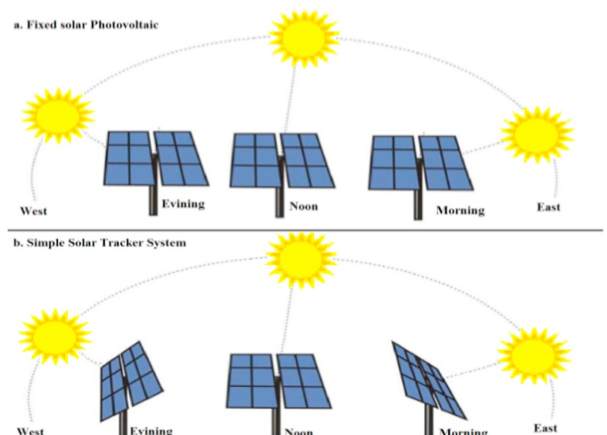


Fig. 2 (a) Without tracking system (b) With tracking system to obtain the maximum solar radiations.

II. Materials and Methods

This project deals with the fabrication of a single axis sun tracker method that electronically controlled the best possible Photo voltaic panel location in accordance with sun. The system consists of solar panel, DC motor, spur gear, sensors (limit switches, light dependent resistor), battery, relays, IC and Arduino microprocessor. Polycrystalline Silicon solar